widthTopological Memory - Dependence

Introduction Today:

Dependence.

Local Stochastic Noise.

Dependence Type I [h] [scale=0.7]

node (u1) at (3,3) [draw,shape=rectangle]u_1; node (x) at (0,6) [draw,shape=circle]x; node (y) at (0,0) [draw,shape=

Pr $[(x,y) \in e|u_1 is satisfied]$
$= \mathbf{Pr}[x \in e|u_1 is satisfied]\,\mathbf{Pr}[y \in e|u_1 is satisfied]$
$\leq \mathbf{Pr}[x \in e]\,\mathbf{Pr}[y \in e]$

Dependence Type II [h] [scale=0.7]

node (u1) at (3,1) [draw,shape=rectangle]u_1; node(u2) at (3,4) [draw,shape=rectangle]u_2; node (x) at (0,6) [draw,s

Pr $[(x,y) \in e|z is not flipped]$
$= \mathbf{Pr}[x \in e|z is not flipped]\,\mathbf{Pr}[y \in e|z is not flipped]$
$\leq \mathbf{Pr}[x \in e]\,\mathbf{Pr}[y \in e]$

widthheightblockDefinition Consider two (qu)bits x, y. We say that z is a **linked node** between x and y if z is eith

Remark Notice that y might be a **linked node** between x and y.

Definition Consider an application of a noise channel. We say that z is a **clean linked node** if:

z is a bit and $z \notin f$ (not flipped).

z is a check and z is satisfied.

. Pr $[x_1, x_2, ..x_k, x_{k+1}, ..x_l \in e|z_1, z_2, ..z_m]$
$= \prod_i \mathbf{Pr}[x_i \in e|z_1, z_2, ..z_m]$
$= \prod_i \mathbf{Pr}[x_i \in e|z_i \in \Gamma(x_i)]$
$\leq \prod_i^k \mathbf{Pr}[x_i \in e] \prod_{i=k+1}^l \mathbf{Pr}[x_i \in e|z_i \in \Gamma(x_i)]$
$\leq M(p)^k$

Local Decoder

definition We say that D is a local decoder if when applied against $2p(1-p)$-depolarized channel:

There exists function $M:(0,1) \to (0,1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.

There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n,k)-Toric, then both the swift-rule and the flip upon majority

widthheightblockDefinition Consider the moment after the application of the depolarizing channel, and right before

For a subset of qubits S, we say that a qubit $x \in S$ is depended if there exists $y \in S$ such:

Either x and y share a check that is not satisfied. (Type I)

Or, x and y have checks which share a bit that was flipped by the application of the channel. (Type II)